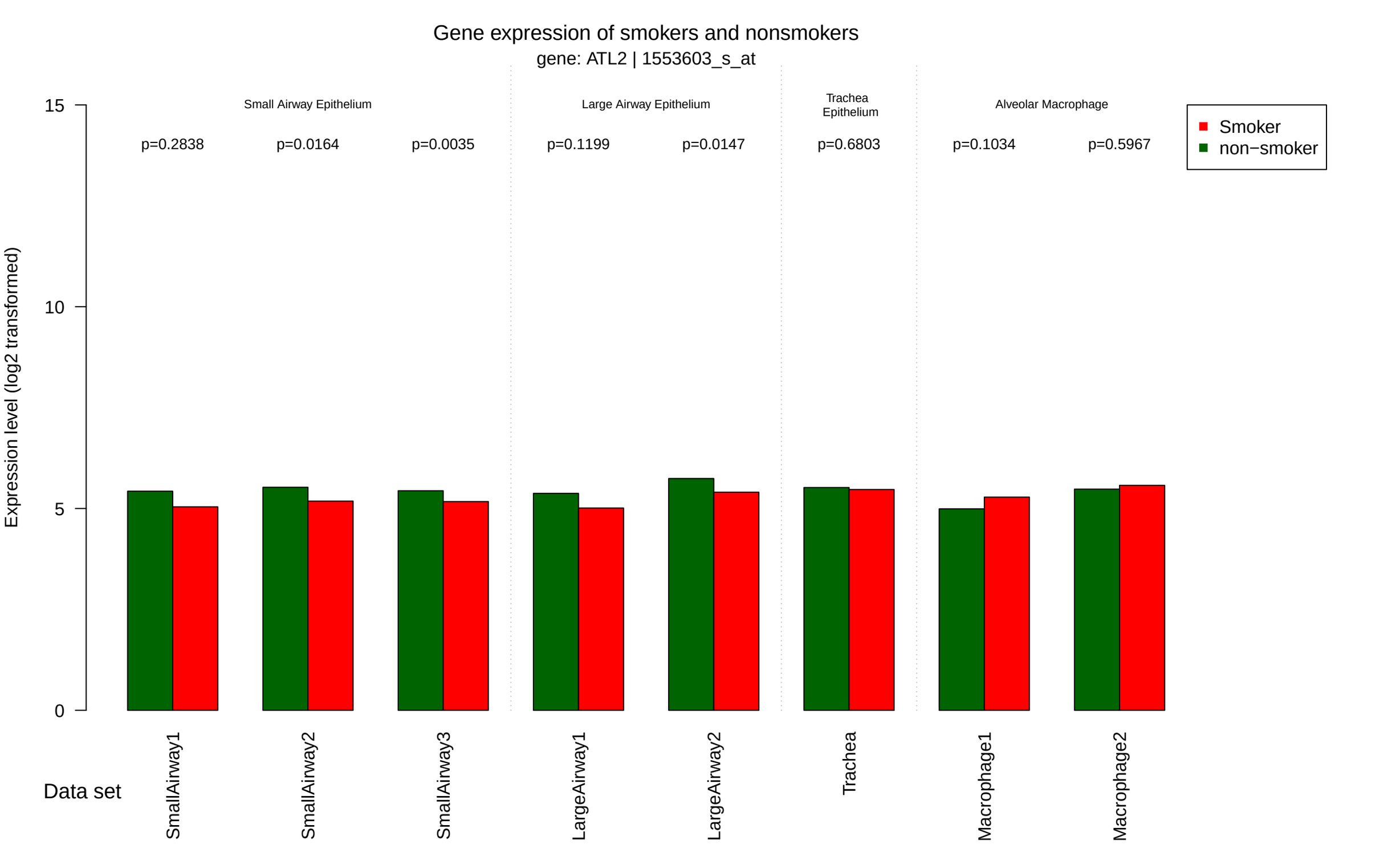
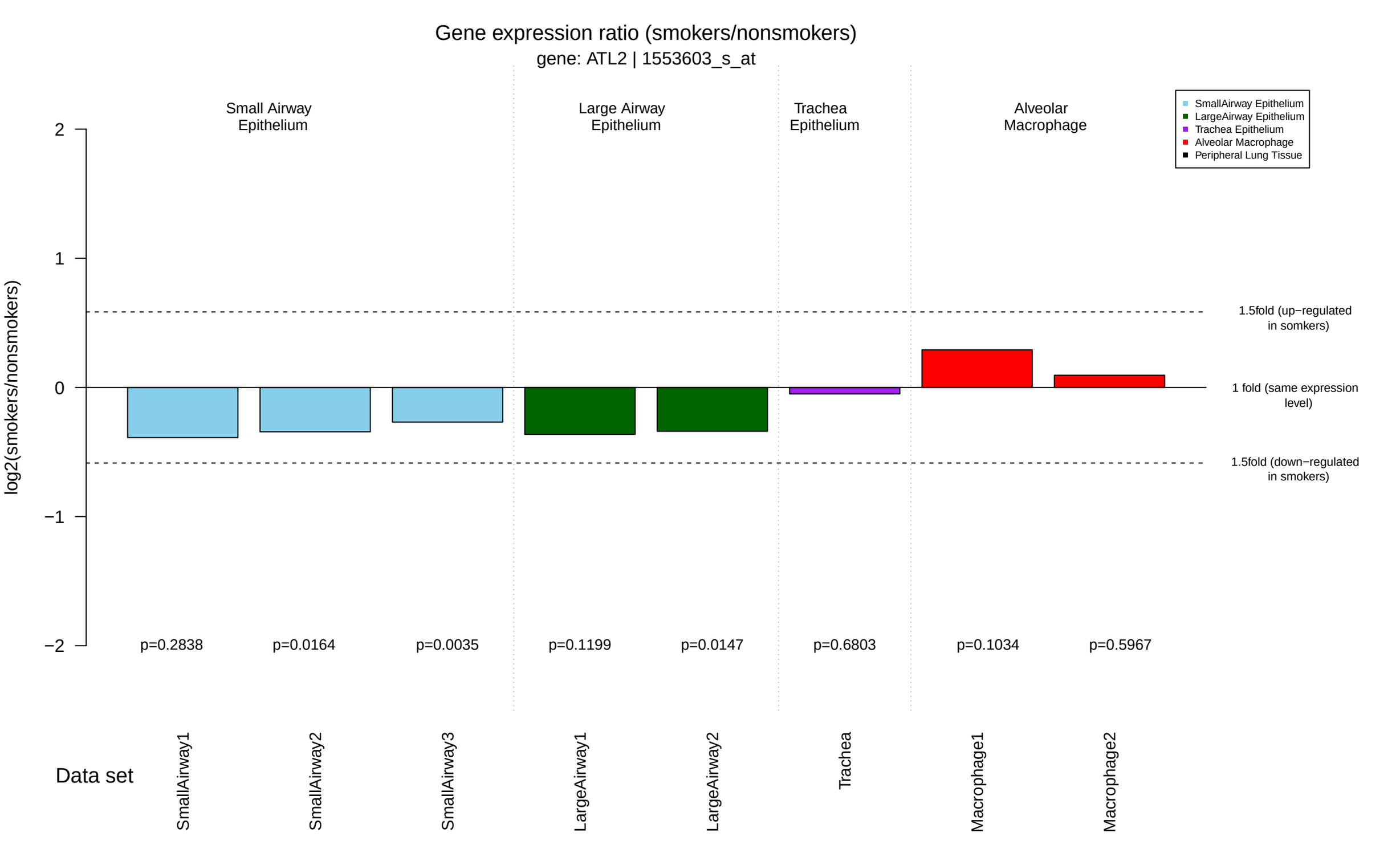
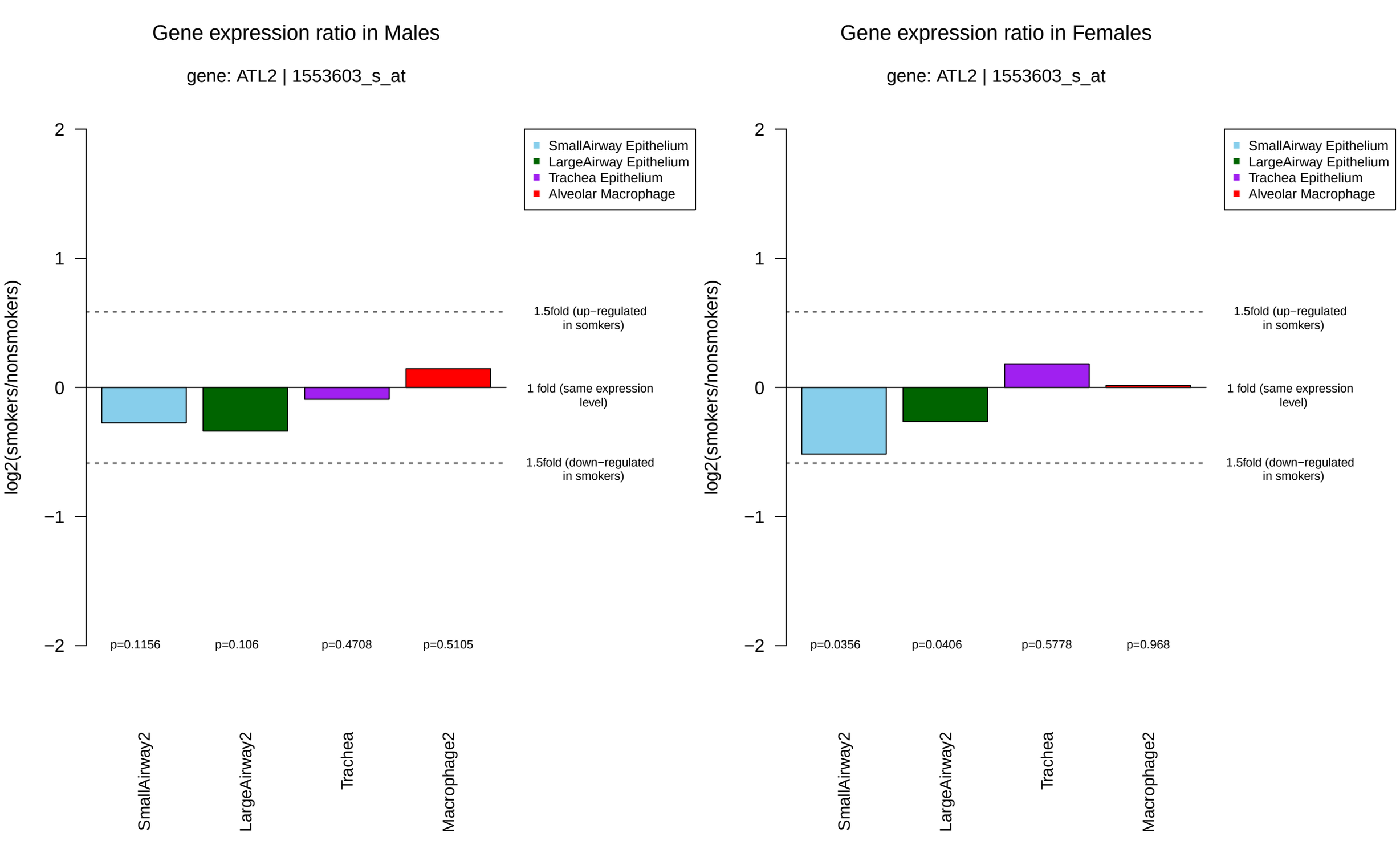
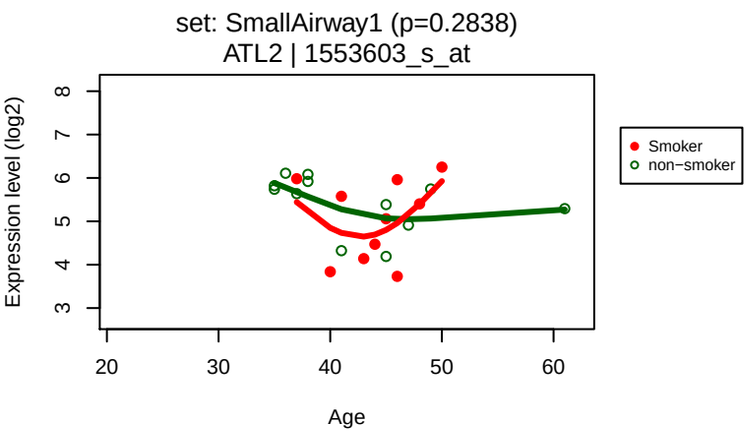
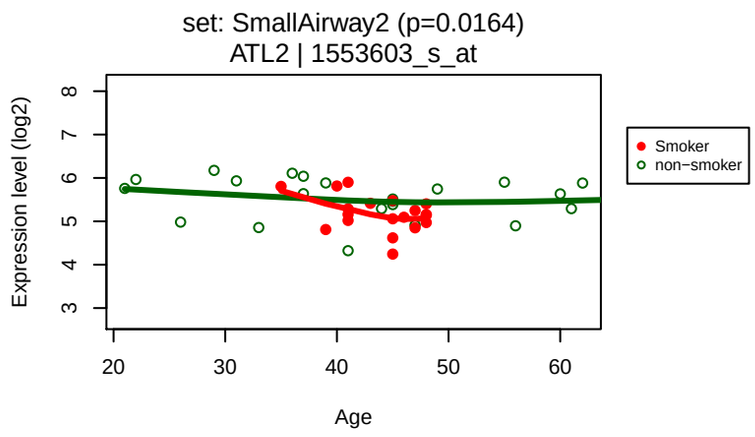
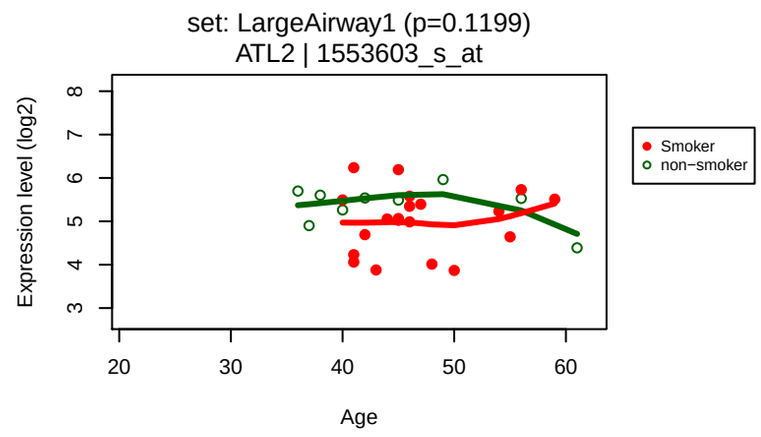

Gene expression of smokers and nonsmokers
gene: ATL2 | 1553603_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2838
p=0.0164
p=0.0035
p=0.1199
p=0.0147
p=0.6803
p=0.1034
p=0.5967
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

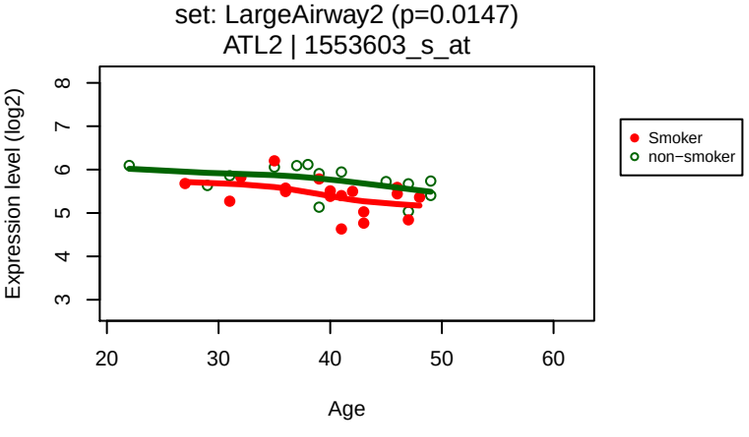

Gene expression ratio (smokers/nonsmokers)
gene: ATL2 | 1553603_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2838
p=0.0164
p=0.0035
p=0.1199
p=0.0147
p=0.6803
p=0.1034
p=0.5967
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

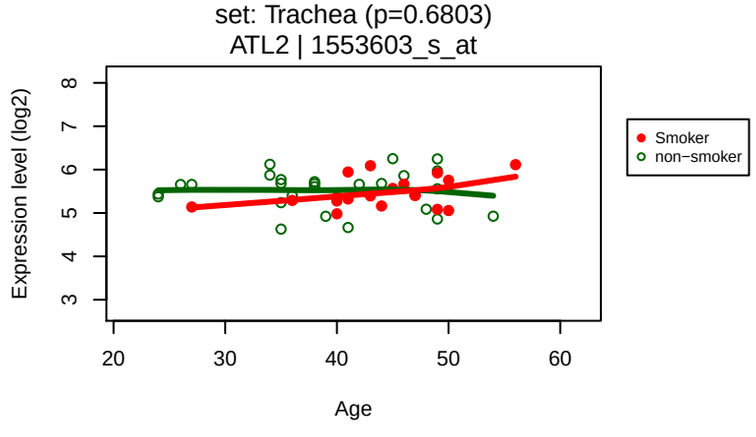

Gene expression ratio in Males
gene: ATL2 | 1553603_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1156
p=0.106
p=0.4708
p=0.5105
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATL2 | 1553603_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0356
p=0.0406
p=0.5778
p=0.968
SmallAirway2
LargeAirway2
Trachea
Macrophage2

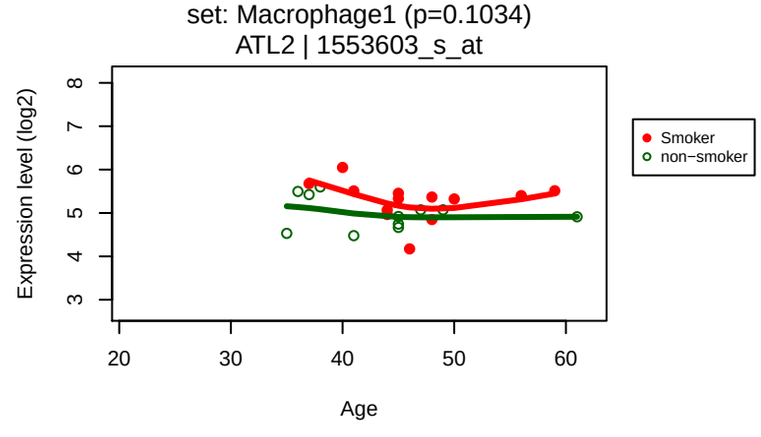

set: SmallAirway1 (p=0.2838)
ATL2 | 1553603_s_at
Expression level (log2)
Age
Smoker
non−smoker

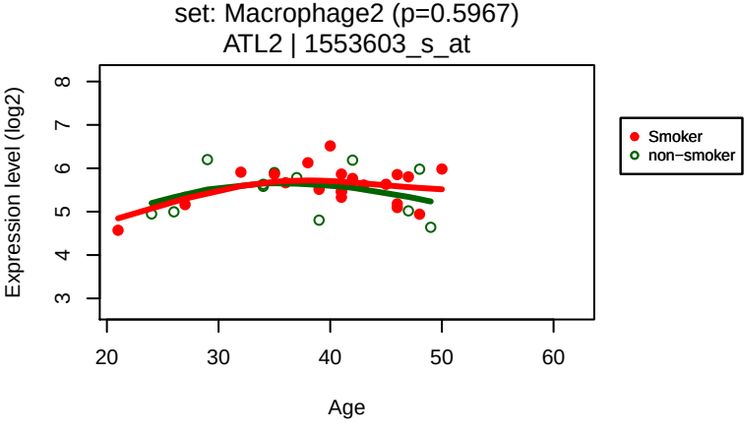

set: SmallAirway2 (p=0.0164)
ATL2 | 1553603_s_at
Expression level (log2)
Age
Smoker
non−smoker

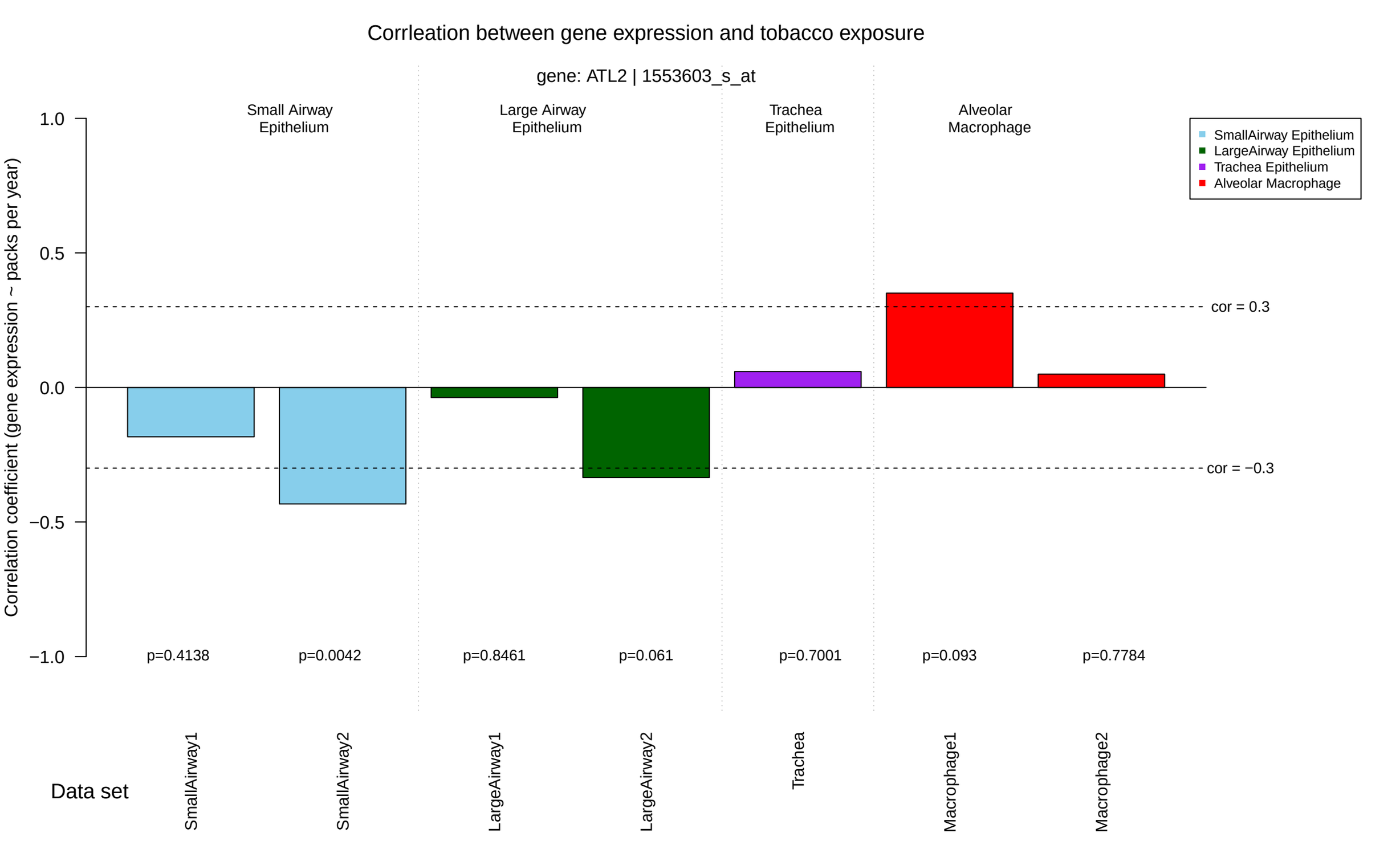

set: LargeAirway1 (p=0.1199)
ATL2 | 1553603_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0147)
ATL2 | 1553603_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6803)
ATL2 | 1553603_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1034)
ATL2 | 1553603_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5967)
ATL2 | 1553603_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATL2 | 1553603_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4138
p=0.0042
p=0.8461
p=0.061
p=0.7001
p=0.093
p=0.7784
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2